

Enterprise Manufacturing Business Intelligence System (EMBIS)

Project Overview

Project Name: Enterprise Manufacturing Business Intelligence System (EMBIS)

Project Type: Business Intelligence Portfolio Project

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Version: 1.0

Completion Date: 14th July 2026

Executive Summary

The Enterprise Manufacturing Business Intelligence System (EMBIS) is an end-to-end Business Intelligence solution developed in Microsoft Excel to transform raw manufacturing data into interactive dashboards that support operational and strategic decision-making.

The solution leverages Power Query, Power Pivot, DAX, PivotTables, PivotCharts, and interactive slicers to automate reporting across multiple business functions.

Business Problem

Manufacturing organizations often manage data across separate operational systems, resulting in fragmented reporting, inconsistent KPIs, and delayed decision-making.

EMBIS addresses this challenge by consolidating data into a centralized reporting platform that delivers accurate, interactive, and executive-ready insights.

Project Objectives

- Design a scalable Business Intelligence reporting solution.
 - Automate data extraction, transformation, and loading (ETL).
 - Implement a Star Schema data model.
 - Develop reusable DAX measures for KPI calculations.
 - Build interactive dashboards for business analysis.
 - Support data-driven decision-making across departments.
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Solution Components

Data Sources

- Sales
- Production
- Inventory

- Payments
- Expenses

ETL

- Power Query

Data Model

- Star Schema
- Fact Tables
- Dimension Tables
- Relationships

Analytics

- DAX Measures
- KPI Engine

Reporting

- PivotTables
- PivotCharts
- Interactive Dashboards
- Slicers

Dashboards Developed

Dashboard	Business Area
Executive Dashboard	Executive Reporting
Sales Dashboard	Sales Performance
Production Dashboard	Manufacturing Operations
Inventory Dashboard	Inventory Management
Finance Dashboard	Financial Performance
Customer Dashboard	Customer Analytics
Supplier Dashboard	Supplier Performance

Technology Stack

Technology	Purpose
Microsoft Excel	Dashboard Platform
Power Query	ETL & Data Transformation
Power Pivot	Data Modelling
DAX	KPI & Business Calculations
PivotTables	Data Aggregation
PivotCharts	Data Visualization
Slicers	Interactive Filtering

Key Features

- Automated ETL process
 - Star Schema data model
 - Centralized DAX calculations
 - Seven interactive dashboards
 - Dynamic KPI cards
 - Interactive slicers
 - Cross-dashboard navigation
 - Standardized dashboard design
 - Executive reporting
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Business Value

EMBIS enables organizations to:

- Monitor enterprise performance in real time.
 - Improve operational visibility across departments.
 - Reduce manual reporting effort.
 - Standardize KPI calculations.
 - Support faster, data-driven decision-making.
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Skills Demonstrated

- Business Intelligence
 - Data Analytics
 - Data Modelling
 - ETL Development
 - Power Query
 - Power Pivot
 - DAX
 - Dashboard Design
 - KPI Development
 - Data Visualization
 - Executive Reporting
 - Microsoft Excel
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Project Deliverables

- Interactive Excel Dashboard Workbook
- Seven Business Intelligence Dashboards
- Data Dictionary
- Entity Relationship Diagram (ERD)
- Data Model Architecture
- KPI Catalogue
- Technical Architecture Document
- Dashboard User Guide
- GitHub Repository
- LinkedIn Portfolio Project

- Power BI Version (*Planned*)

Document Information

Item	Details
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Document	Project Overview
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